Question 1 **10 marks**

Given that the position of an object, as a function of time, in the x direction, is described by the graph below:

(a) What is the *speed* of the object, in the x direction at $t = 0.5$ s?

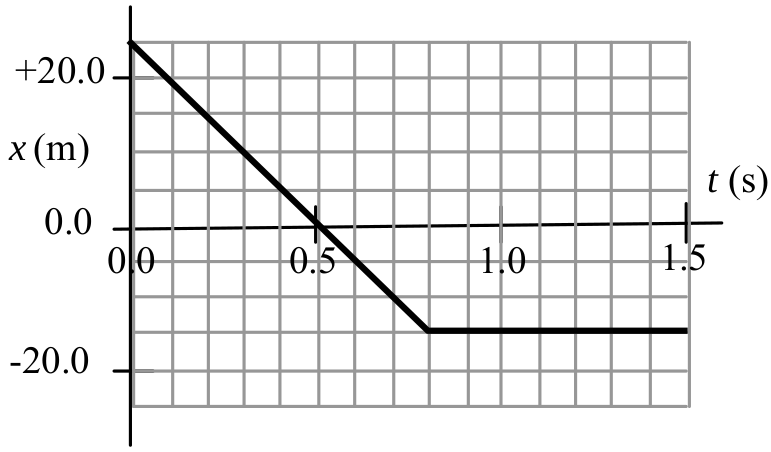
(b) What is the *velocity* of the object, in the x direction at $t = 0.5$ s?

(c) Draw the corresponding velocity time graph.

(d) In what direction is the object travelling at $t = 1.0$ s?

(e) What is the displacement of the object between $t = 0$ s and $t = 1.5$ s?

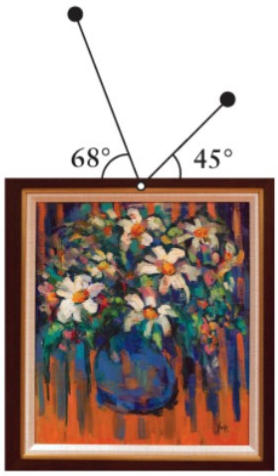
(f) What is the average velocity of the of the object in the from $t = 0$ s to $t = 1.5$ s?



[2 + 1 + 3 + 1 + 1 + 2 = 10 marks]

(a) |v| = 50 m/s
(b) v = -50 m/s
(c) Gradient of position vs time graph: v(t) = -50 m/s from t = 0 s to t = 0.75 s, 0 after
(d) Object is stationary at t = 1.0 s
(e) -35 m
(f) -26.7 m/s

Question 2 **15 marks**



(a) If an object is moving at constant velocity, what can you say about the net force acting on that object?

(b) Consider an example where you are standing on the Earth and holding an apple in your hand. For such an example state two action reaction pairs [both of which must include the apple].

(c) Consider a cricket ball and a tennis ball (each with the same mass). If you drop the tennis ball and cricket ball from the same height (you may ignore air resistance), explain why it hurts more when the cricket ball lands on your foot as compared to the tennis ball.

(d) Describe in words what kinetic (μ_k) and static (μ_s) friction means.

(e) A picture frame is hung using two cords on a wall, as illustrated in the figure to the right. If the mass of the frame is *350* g:

(i) Calculate the tension in each of the cords?

(ii) If the left string is cut what is the tension in the right string?

[2 + 2 + 3 + 2 + (4 + 2) = 15 marks]

(a) Net force = 0

(b) Apple on Earth + Earth on Apple; Sun on Apple, Apple on Sun

(c) F = ma, cricket ball is less elastic than tennis ball, so changes direction quicker when it bounces off your foot: $|a| = \Delta v/\Delta t$, hence the force is greater for the cricket ball than the tennis ball

(d) Refer to lecture notes

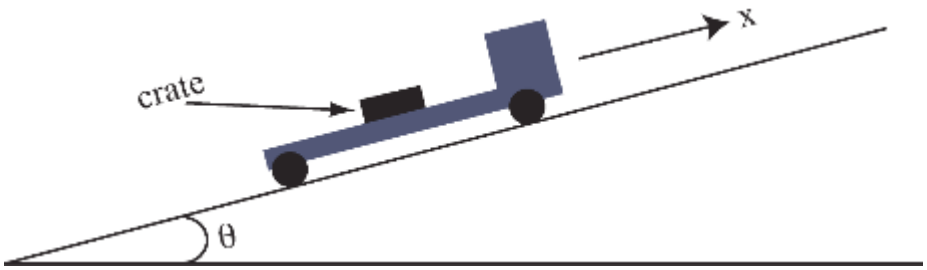
(e)

a. Tension in cord making 68° angle: 2.6N; Tension in cord making 45° angle: 1.4N

b. 3.43N

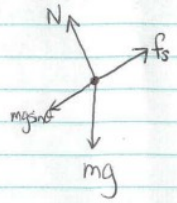
Question 3 **9 marks**

Consider a flatbed truck, carrying a crate up an incline, as illustrated in the figure below.



(a) Draw the force diagram for the crate, when the flatbed truck is stationary, labelling each force.

a)



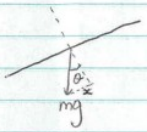
Gravitational force acting on crate pulls it down the slope. Friction acts against motion, so it is up the slope.

(b) Show that the total force acting on the crate, along the x-direction, is

$$F_x = -\,\mathrm{gm}\sin(\theta) + f_s$$

where m is the mass of the crate, g is acceleration due to gravity and f_s is the static frictional force between the crate and the flatbed truck.

b) Component of gravitational force down slope (−ve x-direction)



$x = mg\sin\theta$

~~Component of~~ Friction acting against motion due to gravity

∴ friction up slope (+ve x-direction)

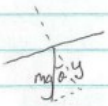
$\therefore F_x = -mg\sin\theta + f_s$

(c) Assuming that the coefficient of static friction is 0.35 and the angle of the incline is 10 degrees calculate the maximum acceleration of the truck before the crate begins to slide.

c) Using Newton's second law,

$F_x = -mg\sin\theta + f_s = ma$

f_s given by: $f_s = \mu N$
$= \mu mg\cos\theta$



$y = mg\cos\theta$

$\therefore ma = -mg\sin\theta + \mu mg\cos\theta$

$a = -g\sin\theta + \mu g\cos\theta$

$= -9.81\sin 10^\circ + 0.35 \cdot 9.81 \cdot \cos 10^\circ$

$= 1.6778\ldots$

$= 1.7\,\mathrm{ms}^{-2}$ up the slope.

Therefore, the truck can accelerate to a max of $1.7\,\mathrm{ms}^{-2}$ up the slope before the crate begins to slide.

[3 + 2 + 4 = 9 marks]

Question 4 **11 marks**

Please provide *brief* answers to the following questions. Approximately one paragraph for each part is sufficient. Diagrams may be used to support your explanations.

(a) According to Newton's third law, when you push on a shopping trolley in the supermarket, the trolley pushes back on you with an equal and opposite force. Explain why it is that you can cause the trolley to move by pushing it, even though these forces are equal and opposite. Your explanation should refer to Newton's laws and all of the forces involved.

(b) A communications satellite occupies a circular geostationary orbit about the earth (that is, it remains fixed above the same point on the earth's equator as the earth rotates). In this orbit, does the satellite experience a force? If so, what is it?

(c) In the sport of Sumo Wrestling, two athletes are placed inside a ring drawn on a floor. To win, one must push the other outside of the ring. Why is it an advantage to be very heavy in this sport?

[4 + 3 + 4 = 11 marks]

a) When you push on the trolley a net force results on the trolley away from you. Then via Newton's second law, this causes the trolley to accelerate away from you. Also via Newton's 3rd law, while you push on the trolley, it pushes back on you with exactly the same force you apply to it, however because you are pushing forward with your feet on the ground via friction, there is a net force on you forward which causes you to accelerate in the forward direction as well.

b) The satellite experiences the force of gravity which causes it to accelerate towards the center of the earth and keeps it in orbit around the planet.

c) A Sumo Wrestler has an advantage if he's heavy because the normal force between his feet and the floor is large and the friction force between his feet and the floor is larger. This makes it hard to push him out of the ring.

Question 5 **8 marks**

Please provide *brief* answers to the following questions. Approximately one paragraph for each part is sufficient.

(a) Imagine you are an astronaut standing on the moon and you simultaneously drop a hammer and feather? Which hits the surface of the moon first? The Hammer, the feather or do they both hit at the same time? Explain your reasoning.

(b) Now imagine you're standing on the Earth and drop the hammer and feather at the same time. Which hits the surface of the Earth first? The Hammer, the feather or do they both hit at the same time? Explain your reasoning.

(c) A book is placed on a desk. A second book is placed on top of it doubling the force due to gravity on the desk. What happens to the atoms of the desk that enables the normal force from the desk to provide exactly the force equal and opposite the two books to prevent any acceleration?

[2 + 3 + 3 = 8 marks]

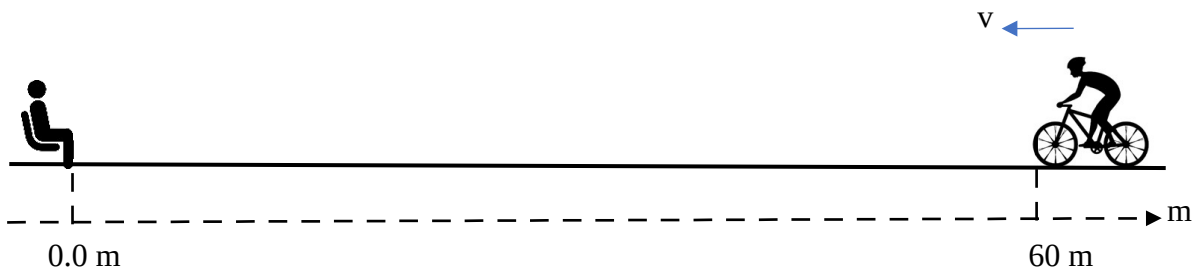
(a) I would expect the hammer and feature to hit the ground simultaneously on the moon since the gravitational force on each is directly proportional to their mass. Consequently they accelerate at the same rate.

(b) On the Earth I would expect the hammer to hit the ground first. Although the gravitational force would cause both object to accelerate at the same rate in a vacuum, on Earth the effect of air resistance is much greater on the feather causing it to reach terminal velocity well before the hammer.

(c) The books cause the atoms in the desk to be shifted from their mean position. As they shift, they push back until the force pushing them is exactly balanced and no further distortion occurs. The net result is the normal force which exactly opposite the force of the books on the table.

Question 6 **12 marks**

You're sitting on a bench watching a cyclist riding directly towards you. You are equipped with a radar gun which enables you to accurately measure the distance of the cyclist and his velocity. When he is 60 m distant, you observe he is riding towards you at -4.0 ms^{-1}. Then 2.0 seconds later you measure his velocity as -7.0 ms^{-1}.



(a) Calculate his acceleration over this 2.0 s period.
(b) Assuming his acceleration is constant, calculate how far away from you is he after this 2.0 s.
(c) The cyclist then travels towards you at the constant velocity of $-7.0\,ms^{-1}$. Calculate how far away from you he is after a further 3.0 seconds.
(d) The cyclist then sees you and puts on his brakes to stop before hitting you. Calculate the acceleration required to bring him to rest before hitting you.

[2 + 4 + 2 + 4 = 12 marks]

(a) $a=\frac{v_{f-v_i}}{\Delta t}=\frac{-7.0-(-4.0)}{2.0}=\frac{-7.0+4.0}{2.0}=-1.5\,m\,s^{-2}$

(b) $2a\Delta x=v_f^2-v_i^2$ hence $2\times(-1.5)\times\Delta x=49-16=33$

$\Delta x=\frac{-33}{3.0}$= -11.0 m

Distance = 60 $+\Delta x=60-11=49\,m$

(c) $\Delta x=49+-7.0\times 3$
$=49-21=28\,m$

(d) $2a\Delta x=v_f^2-v_i^2$

$2a\times(-28)=-49$

$a=\frac{-49}{-2\times 28}=0.875\,m\,s^{-2}$

Question 7 **20 marks**

Tegan Philip is the goal attack of the 2017 Melbourne Vixens's netball team. We shall analyze how Tegan shoots a goal from a distance 4.800 metres to the ring. This situation is shown in figure 1 below.

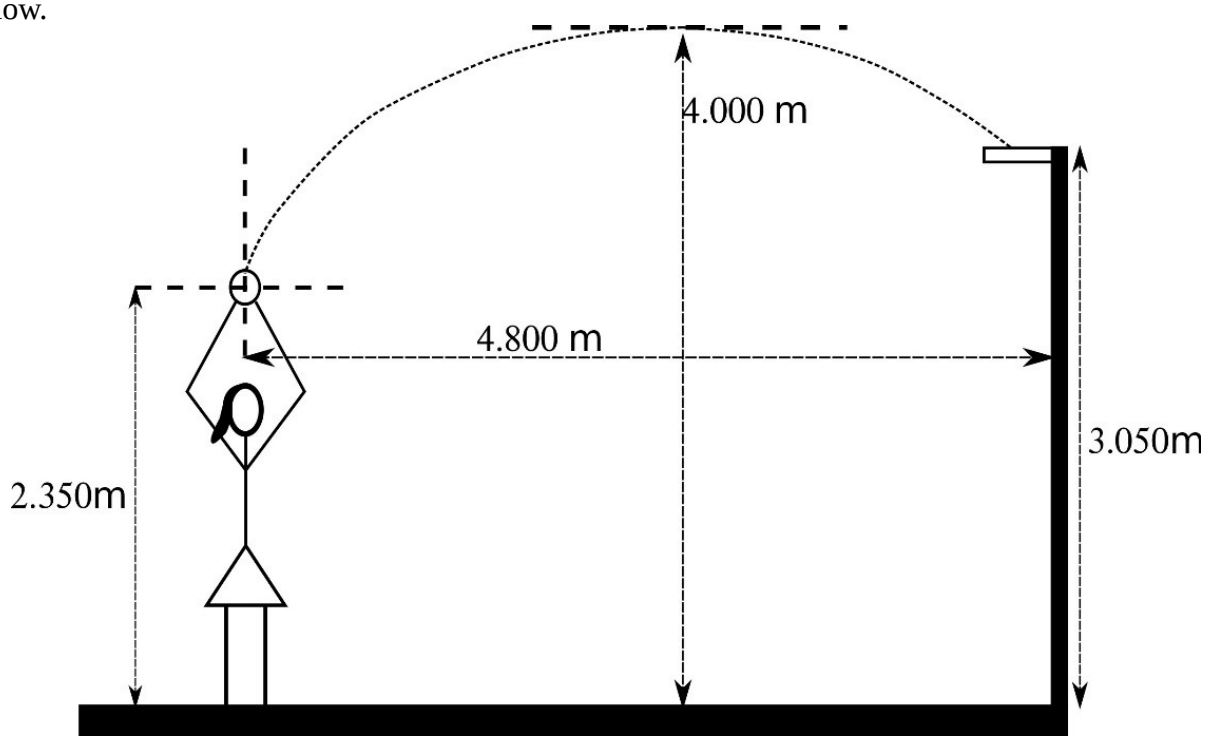


Figure 1. Tegan Philip shoots a goal 4.800 m from the ring. The netball ring is 3.050 m from the ground. The maximum height of the ball during its flight is 4.000 m. Tegan stands still, throws and releases the ball from a height of 2.350 m from the ground. Assume the acceleration due to gravity is 9.800 $m\,s^{-2}$.

a) Write down the equations for the location of the ball in the horizontal x and vertical y dimensions as a function of time t. Use expressions including the initial horizontal velocity of ball v_{ix}, the initial vertical velocity of the ball v_{iy}, and the acceleration due to gravity g.
b) Use a constant acceleration formula to find the initial vertical velocity v_{iy} required for the ball to reach the maximum height of the trajectory 4.000 m from the ground.
c) Use a constant acceleration formula to find the time from Tegan releasing the ball until it reaches its maximum height 4.000 m from the ground.
d) Calculate the time for the ball to fall from the maximum height 4.000 m to the height of the ring located 3.050 m from the ground.
e) Calculate the total time for the ball to travel up to 4.000 m and fall back to 3.050 m.
f) Calculate the initial horizontal velocity v_{ix} required to make the ball follow the trajectory shown and land in the ring.

[2 + 4 + 4 + 3 + 2 + 5 = 20 marks]

(a) $x(t)=x_i+v_{ix}t,\ y(t)=y_i+v_{iy}t-0.5gt^2$

(b) $v_f^2-v_i^2=2a\Delta x$ so $0-v_i^2=-2g(4.000-2.350)=-32.34$ so $v_i=5.687\,ms^{-1}$

(c) $v(t)=v_{iy}-gt\Rightarrow 0=5.687-9.800t\Rightarrow t=\frac{5.687}{9.800}=0.580\,s$

(d) $y(t)=4.000-0.5gt^2=3.050\Rightarrow t^2=\frac{4.000-3.050}{0.5\times 9.800}\Rightarrow t=0.440\,s$

(e) t = 0.580 + 0.440 = 1.020 s

(f) $x(t)=x_i+v_{ix}t=4.800=0+v_{ix}t\Rightarrow v_{ix}=\frac{4.800}{1.020}=4.704\,ms^{-1}$

Question 8 **25 marks**

A 50 cent coin of mass 15.5 grams slides down a slope and then continues across a horizontal surface before stopping a distance L from the end of the slope (see figure 2). The coefficient of kinetic friction between the coin and both surfaces is $\mu_K=0.60$.

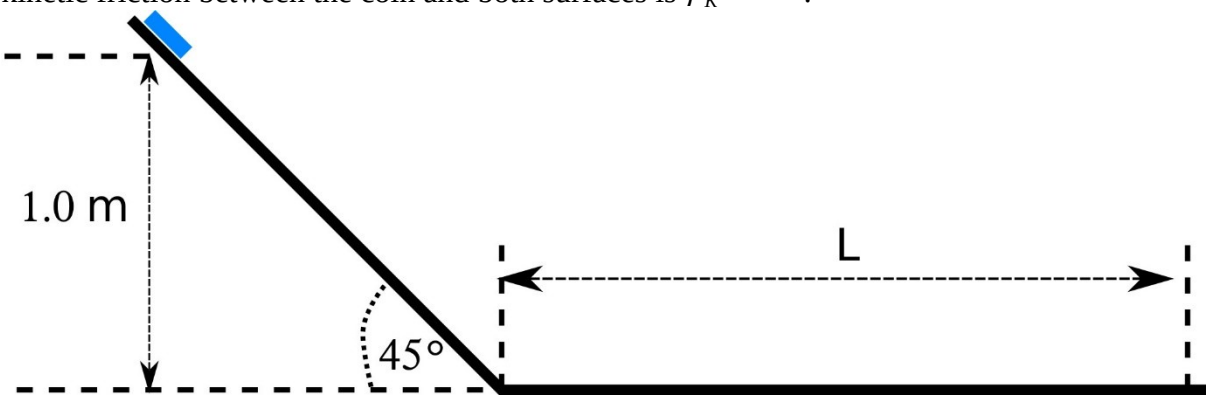


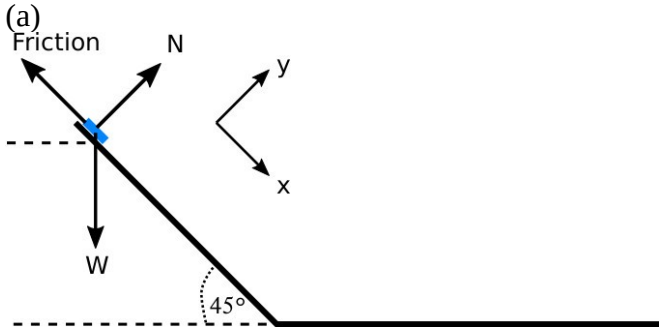
Figure 2. A 50 cent coin slides down a straight surface angled at 45 degrees to the horizontal before coming to rest a distance L beyond the end of the slope. The coefficient of kinetic friction is $\mu_K=0.60$. The coins starts with an initial velocity of 0.50 ms^{-1}.
The initial height of the coin above the surface is 1.00 metres. The initial velocity of the coin is 0.50 ms^{-1}.

(a) Sketch a diagram in your exam booklet showing all the forces acting on the coin at the top of the slope. Include the coordinate axes you'll use to solve the problem.
(b) Write down Newton's laws of motion for the x and y coordinates you've chosen to solve the problem.

Calculate:
(c) the force due to the kinetic friction as it slides down the slope,
(d) the acceleration of the coin as it slides down the slope,
(e) the speed of the coin at the end of the slope,
(f) the kinetic energy of the coin at the bottom of the slope,
(g) the force due to kinetic friction as it slides across the horizontal surface,
(h) how much work must be done to the coin to stop its motion,
(i) how far the coin slides across the flat surface before stopping.

[5 + 2 + 2 + 4 + 4 + 2 + 2 + 2 + 2 = 25 marks]

(a)



(b)

$Wsin(45) - N = m\,a_y$

$Wcos(45) - F_s = m\,a_x$

(c)

$\sin(45) - N = m\,a_y \Rightarrow N = mgsin(45)$

$F_k = \mu_K N = 0.60 \times 0.0155 \times 9.8 \times \sin(45) = 0.0644\ Newtons$

(d)

$Wsin(45) - N = m\,a_y \Rightarrow N = mgsin(45)$

$Wcos(45) - F_s = m\,a_x$

$\Rightarrow mgcos(45) - \mu_k N = ma$

$mgcos(45) - 0.6\,mgsin(45) = m\,a_x \Rightarrow a_x = \frac{mg\,0.707(1-0.6)}{m} = 9.8 \times 0.707 \times 0.4$

$\Rightarrow a_x = 2.8\,m\,s^{-2}$

(e) $v_f^2 - v_i^2 = 2\,a_x\,\Delta x \Rightarrow v_f^2 = 2 \times 2.8 \times \frac{1.0}{\cos(45)} + 0.5^2 = 8.17 \Rightarrow v_f = 2.86\,m\,s^{-1}$

(f) $KE = \frac{1}{2} m\,v_f^2 = 0.5 \times 0.0155 \times 8.17 = 0.0633\,J$

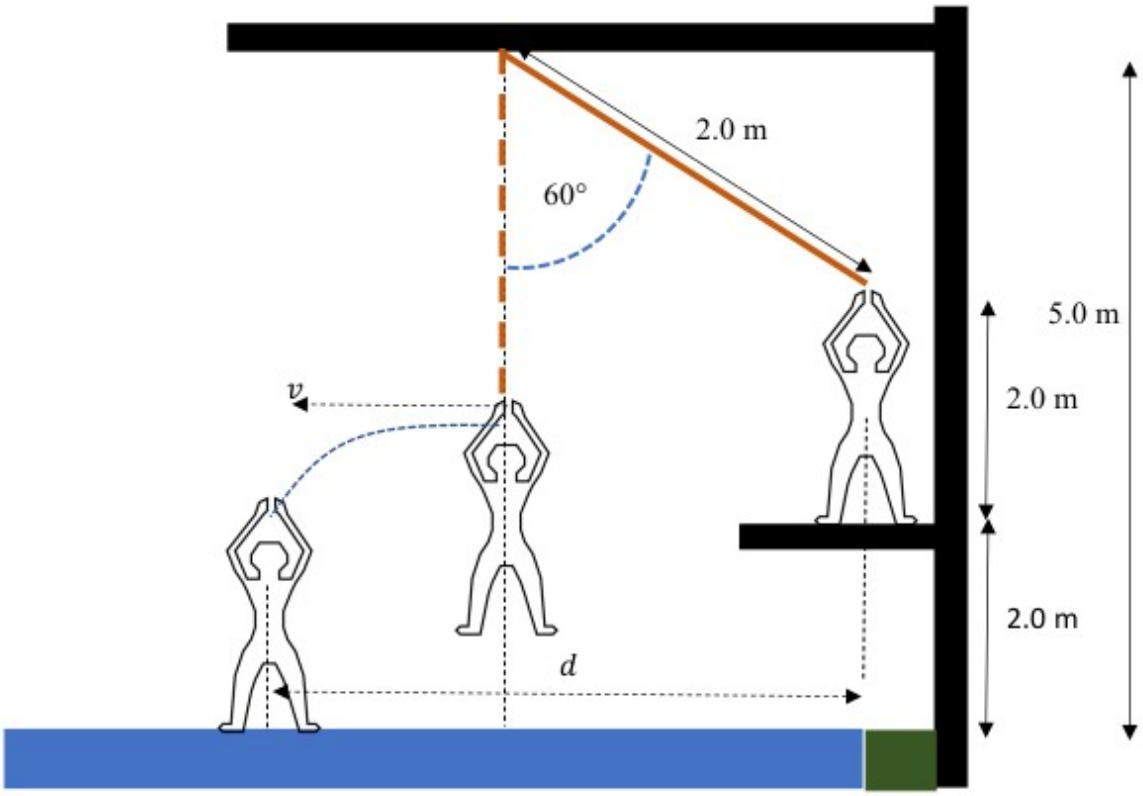
(g) $F = \mu_k mg = 0.6 \times 0.0155 \times 9.8 = 0.091\ Newtons$

(h) Work from friction = KE = 0.0633J

$$\text{Work} = F \times d = 0.0633 \Rightarrow d = \frac{0.0633}{0.091} = 0.696\,m$$

Question 9 **30 marks**

Sophie plays with a rope attached to a tree branch hanging above a lake as shown in the figure below. Sophie has a mass of 50 kg. In all the calculations that follow, until she releases her hold on the rope, assume Sophie holds the rope at a distance 2.0 m above her feet. Neglect air resistance and friction. Use $g = 9.8\,ms^{-2}$.



(a) Sophie stands on a branch of the tree directly above the edge of the lake. She steps off the branch and swings out over the lake. Using the principle of conservation of energy, calculate Sophie's horizontal velocity v when the rope is directly vertical (i.e. at the bottom of the swing). Assume she begins her swing with an initial speed of zero.
(b) Calculate Sophie's acceleration (direction and magnitude) at this point.
(c) Draw a force diagram of Sophie and the swing and calculate the tension in the rope at this point.

Sophie releases the rope at this point and follows a projectile motion until her feet touch the lake. Assume that Sophie keeps the same body posture in the calculations that follow.

(d) Calculate the horizontal distance from the edge of the lake when she releases the rope.

(e) Calculate the time between Sophie releasing the rope and her feet hitting the water of the lake.
(f) Calculate the distance, d, from the edge of lake to where Sophie's feet hit the water.
(g) Calculate Sophie's total speed when her feet hit the water and hence her kinetic energy.

[6 + 4 + 4 + 3 + 4 + 3 + 6 = 30 marks]

(a) At bottom of swing change in height = 1.0m
Gravitation PE change = $mgh = 50 \times 9.8 \times 1.0 = 490.5\,J$

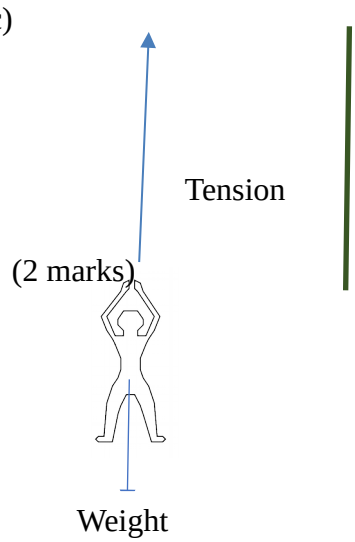
K.E. = $\frac{1}{2} m v^2 = 490.5$

=> $v = \sqrt{\frac{2 \times 490.5}{50}} = 4.43\,m\,s^{-1}$

(b) Sophie is accelerating directly upwards due to the tension in the rope as it it swings here in a circle centered on the tree branch

$a = \frac{v^2}{r} = \frac{4.43^2}{2} = 9.8\,m\,s^{-2}$

(c)



(2 marks)

Net force = Tension – Weight = ma =>
Tension = ma+mg =
$50 \times 9.8 + 50 \times 9.8 = 980\,N$

(d) Distance to lake edge = $2.0 \times \sin(60) = 1.0\,m$

(e) $\Delta x = v_i + \frac{1}{2} a (\Delta t)^2$ => $-1.0 = 0.0 - \frac{1}{2} 9.8 (\Delta t)^2 \Rightarrow \Delta t = \sqrt{\frac{2.0}{9.8}} = 0.45\,s$

(f) d=$1.0 + 4.43 \times 0.45 = 3.0\,m$

(g) Final vertical velocity = $a \times \Delta t = 9.8 \times 0.45 = 4.41\,m\,s^{-1}$

Total speed = $\sqrt{v_x^2 + v_y^2} = \sqrt{4.43^2 + 4.41^2} = 6.3\,m\,s^{-1}$

KE = $\frac{1}{2} m v^2 = \frac{1}{2} 50 \times (6.3)^2 = 980\,J$